The Unit Circle and Exact Values

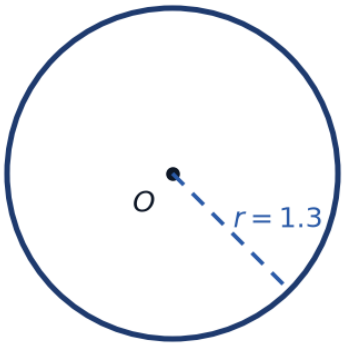
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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Points on the unit circle**

The terminal side of angle θ in standard position passes through the point $P\left(-\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle.

Find:



a $\sin\theta$

b $\cos\theta$

c $\tan\theta$

d the quadrant containing θ

2 **[Foundational]** • **Target: Points on the unit circle**

State the reference angle for each angle:

a $\frac{5\pi}{6}$

b $\frac{4\pi}{3}$

c $-\frac{\pi}{4}$

d $\frac{7\pi}{2}$

3 **[Foundational]** • **Target: Points on the unit circle**

Identify the quadrant(s) in which each pair of conditions holds:

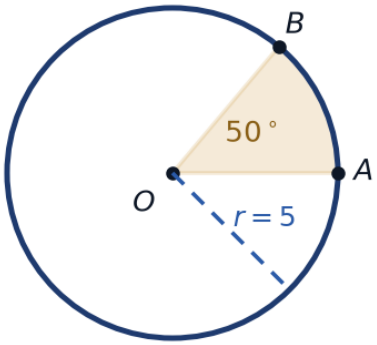
a $\sin\theta < 0$ and $\cos\theta > 0$

b $\tan\theta > 0$ and $\sin\theta < 0$

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4 [Foundational] • Target: Points on the unit circle

Given that $\cos\theta = -\frac{5}{13}$ and θ is in Quadrant III, find $\sin\theta$ and $\tan\theta$ exactly.



5 [Foundational] • Target: Even/odd properties and periodicity

Use the fact that cosine is even and sine is odd to simplify each expression:

a $\sin(-\theta) + \sin\theta$

b $\cos(-\theta) - \cos\theta$

c $\frac{\sin(-\theta)}{\cos(-\theta)}$

6 [Foundational] • Target: Even/odd properties and periodicity

Use periodicity to find the exact value of each expression:

a $\sin\frac{13\pi}{6}$ **b** $\cos\frac{17\pi}{4}$ **c** $\tan\frac{9\pi}{4}$

7 [Foundational] • Target: Solving basic trigonometric equations on the unit circle

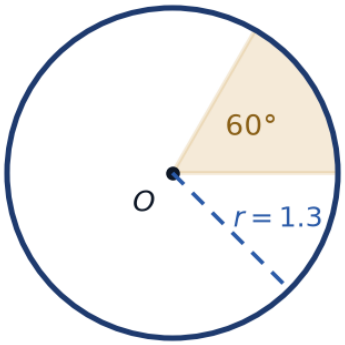
Find all solutions to $\sin\theta = -\frac{1}{2}$ in the interval $[0, 2\pi)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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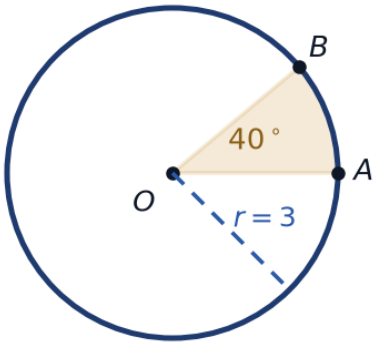
8 [Intermediate] • Target: Building the unit circle from special triangles

Using the 30-60-90 triangle, derive the exact coordinates of the point on the unit circle at $\theta = \frac{\pi}{3}$.



9 [Intermediate] • Target: Building the unit circle from special triangles

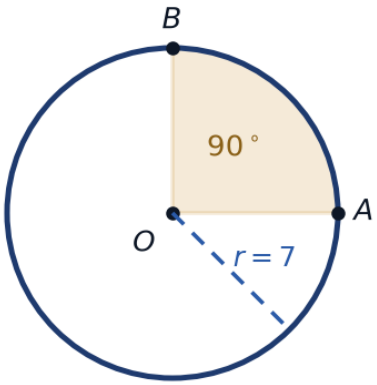
State the coordinates of all four points on the unit circle where the terminal side lies along an axis ($\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$).



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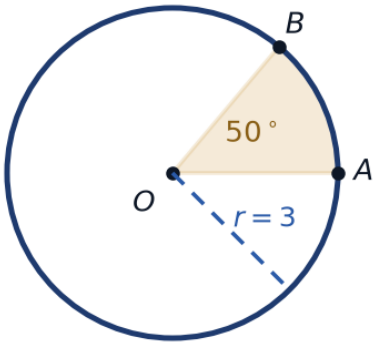
10 [Intermediate] • Target: Using the Pythagorean identity with unit circle points

A point on the unit circle has x-coordinate $\frac{7}{25}$ and lies in Quadrant IV. Find its y-coordinate using $x^2 + y^2 = 1$.



11 [Intermediate] • Target: Using the Pythagorean identity with unit circle points

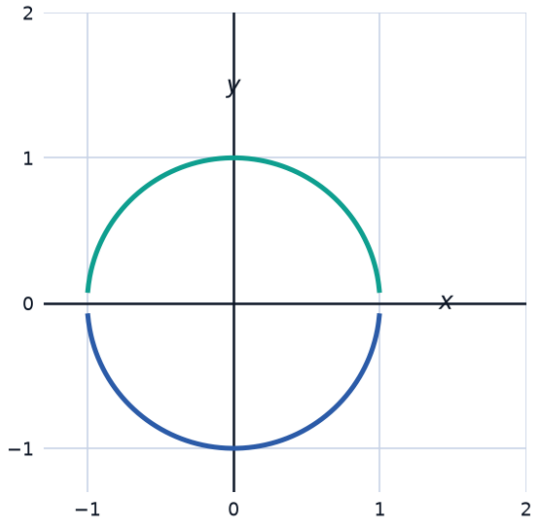
Given $\sin\theta = 0.6$ and θ in Quadrant II, find $\cos\theta$ and $\tan\theta$ exactly (using the fact $0.6 = \frac{3}{5}$).



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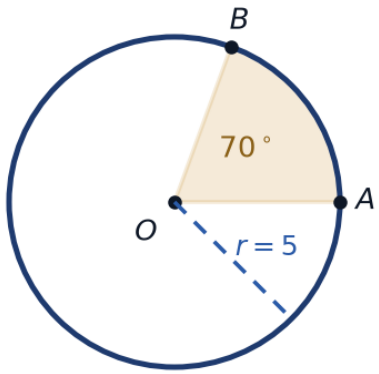
12 [Intermediate] • Target: The unit circle picture

The unit circle $x^2 + y^2 = 1$ is shown. Use it to estimate $\cos\left(\frac{2\pi}{3}\right)$ and $\sin\left(\frac{2\pi}{3}\right)$ visually, then state the exact values.



13 [Intermediate] • Target: All six trig functions from a unit circle point

A point on the unit circle has coordinates $\left(-\frac{5}{13}, \frac{12}{13}\right)$. Find all six trigonometric function values for the corresponding angle θ.



14 [Intermediate] • Target: Solving for the angle given exact coordinates

Find all values of θ in $[0, 2\pi)$ for which the unit circle point is exactly $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Using the Pythagorean identity with unit circle points • Cross-Lesson: 7.04 Applications: Growth and Decay

Level 1 · FoundationsFind the exact value of $\tan\left(\frac{11\pi}{6}\right)$ and $\sec\left(\frac{11\pi}{6}\right)$ using reference angles and unit circle coordinates.

16 [Challenging] • Target: The unit circle picture • Cross-Lesson: 8.01 Rational Functions and Asymptotes

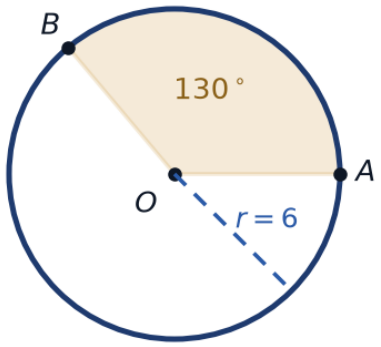
Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

17 [Challenging] • Target: All six trig functions from a unit circle point • Cross-Lesson: 8.02 Graphing Rational Functions

Level 3 · AdvancedFind all values of θ in $[0, 2\pi)$ for which the rose curve $r = 4\sin(3\theta)$ passes through the pole (origin), and find the maximum distance of any point on the curve from the origin.

18 [Challenging] • Target: Solving for the angle given exact coordinates • Cross-Lesson: 8.03 Introduction to Limits

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



19 [Challenging] • Target: Exact values • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

20 [Challenging] • Target: Points on the unit circle • Cross-Lesson: 9.02 Geometric Sequences and Series

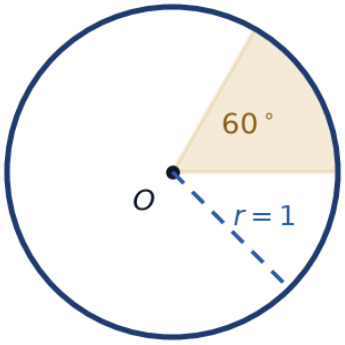
Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exact values

A point P is on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. The coordinates of P are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. If a point Q on the unit circle has an angle ϕ such that $\phi = \theta + \frac{\pi}{3}$, what is the y-coordinate of point Q?



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

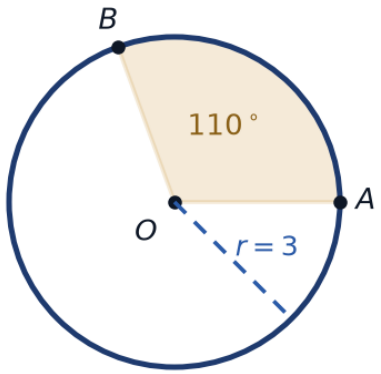
c. 1

d. $\frac{\sqrt{2}}{2}$

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22 [Foundational] • Target: Points on the unit circle

The point P is located on the unit circle at an angle of $\frac{\pi}{3}$ radians counterclockwise from the positive x-axis. What are the Cartesian coordinates (x, y) of point P?



a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

b. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

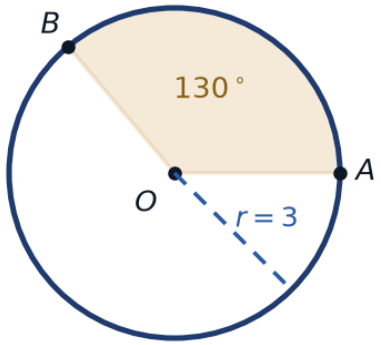
c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

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23 [Foundational] • Target: Even/odd properties and periodicity

A point P is located on the unit circle such that its angle with the positive x-axis is $\frac{2\pi}{3}$ radians. What are the coordinates of point P?



a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

24 [Foundational] • Target: Solving basic trigonometric equations on the unit circle

A circle is centered at the origin, and a point P lies on the circle. The angle formed by the positive x-axis and the line segment OP (where O is the origin) is θ. The coordinates of point P are $(r\cos\theta, r\sin\theta)$. If point P has coordinates $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$ and lies on a circle of radius R, find the value of R.

a. $\frac{5}{2}$

b. 5

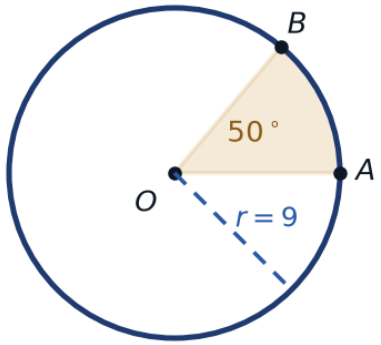
c. $\frac{5\sqrt{3}}{2}$

d. 10

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25 [Foundational] • Target: Building the unit circle from special triangles

A point P lies on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. If the x-coordinate of P is $\frac{1}{2}$, what is the y-coordinate of P? Assume θ is in the first quadrant.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{1}{2}$

d. $\sqrt{3}$

26 [Foundational] • Target: Using the Pythagorean identity with unit circle points

If $\sin\theta = \frac{3}{5}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{37}{50}$

b. $\frac{16}{25}$

c. $\frac{4}{5}$

d. $\frac{9}{25}$

27 [Foundational] • Target: The unit circle picture

A point P has polar coordinates (r, θ). When converted to Cartesian coordinates, P is $(3, 3\sqrt{3})$. If the angle θ is restricted to the interval $[0, 2\pi)$, find the value of r and θ in radians.

a. $r = 6, \theta = \frac{\pi}{3}$

b. $r = 3\sqrt{2}, \theta = \frac{\pi}{4}$

c. $r = 6, \theta = \frac{\pi}{6}$

d. $r = 3, \theta = \frac{\pi}{3}$

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28 [Foundational] • Target: All six trig functions from a unit circle point

A particle's position in the Cartesian plane is described by the polar coordinates (r, θ), where r is the distance from the origin and θ is the angle measured counterclockwise from the positive x-axis. The particle begins at $\theta_1 = \frac{\pi}{6}$ with a radial distance of $r_1 = 4$. It then moves to a new position such that its new angle is $\theta_2 = \frac{2\pi}{3}$ and the radial distance is $r_2 = 6$. If the displacement vector of the particle is $\mathbf{d} = \langle x, y \rangle$, what is the value of $x^2 + y^2$?

a. 13

b. 25

c. 49

d. 52

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Solving for the angle given exact coordinates

Consider a right-angled triangle ABC, where the right angle is at vertex B. Point D lies on the hypotenuse AC such that BD is perpendicular to AC. If AB = 10 units and BC = 24 units, find the length of AD.

a. $\frac{100}{26}$

b. $\frac{240}{26}$

c. $\frac{576}{26}$

d. $\frac{600}{26}$

30 [Intermediate] • Target: Exact values

The terminal ray of an angle θ in standard position in the xy-plane passes through the point $(-5, 12)$. What is the value of $\sin\theta$?

a. $-\frac{5}{13}$

b. $\frac{12}{13}$

c. $-\frac{12}{5}$

d. $\frac{12}{5}$

31 [Intermediate] • Target: Points on the unit circle

Prove the identity $\sin 3A = 3\sin A - 4\sin^3 A$ using the angle addition formula.

a. Not an identity

b. Only for $A < \pi/2$

c. Use calculus

d. $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓

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32 [Intermediate] • Target: Even/odd properties and periodicity

Pythagorean identity: $1 + \tan^2\theta =$

a. $\csc^2\theta$

b. $\sec^2\theta$

c. $\cot^2\theta$

d. $\cos^2\theta$

33 [Intermediate] • Target: Solving basic trigonometric equations on the unit circle

A circle has radius 5. Find the arc length subtended by a central angle of $\frac{1\pi}{2}$ radians.

a. $5/4\pi$

b. $5/2\pi$

c. $7/2\pi$

d. 5π

34 [Intermediate] • Target: Building the unit circle from special triangles

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

a. 5

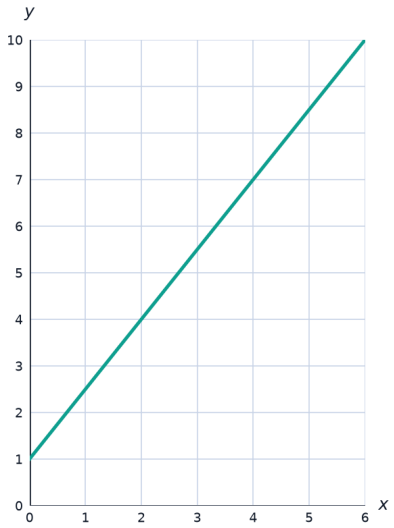
b. 7

c. 25

d. 16

35 [Intermediate] • Target: Using the Pythagorean identity with unit circle points

In polar form, which equation represents a circle of radius 3 centred at the origin?



a. $r = 3$

b. $r = 3\theta$

c. $r = 3\cos\theta$

d. $\theta = 3$

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36 [Intermediate] • Target: The unit circle picture

Find θ (in $[0^\circ, 360^\circ)$) for the Cartesian point $(3, 0)$.

a. 180

b. 0

c. 181

d. 90

37 [Intermediate] • Target: All six trig functions from a unit circle point

Find r for the Cartesian point $(3, 4)$.

a. 7

b. 3

c. 4

d. 5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving for the angle given exact coordinates • Cross-Lesson: 8.03 Introduction to Limits

Relative to a fixed origin O, the lines L_1 and L_2 have vector equations $L_1 : \mathbf{r} = (1; 0; -1) + \lambda(1; 2; 2)$, $L_2 : \mathbf{r} = (2; -4; 7) + \mu(2; 1; -2)$. The unique points P on L_1 and Q on L_2 are such that the line segment PQ is perpendicular to both L_1 and L_2. A point R lies on L_1 such that the area of triangle PQR is 18. Given that the z-coordinate of R is positive, what is the exact distance from the origin O to the midpoint of the line segment QR?

a. $3\sqrt{5}$

b. $\sqrt{13}$

c. $6\sqrt{2}$

d. $\sqrt{77}$

39 [Challenging] • Target: Exact values • Cross-Lesson: 9.01 Arithmetic Sequences and Series

A curve passes through the point $\left(1, \frac{1}{2}\right)$ and satisfies the differential equation $\frac{dy}{dx} = \frac{y^2(1 - \ln x)}{x^2}$ for $x > 0$. Determine the exact coordinates of the stationary point of the curve and its nature.

a. $\left(e, \frac{e}{2e-1}\right)$, which is a local maximum

b. $\left(e, \frac{1}{2e-1}\right)$, which is a local maximum

c. $\left(e, \frac{e}{2e+1}\right)$, which is a local minimum

d. $\left(\frac{1}{e}, \frac{1}{2e+1}\right)$, which is a local minimum

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40 [Challenging] • Target: Points on the unit circle • Cross-Lesson: 9.02 Geometric Sequences and Series

Consider the exponential function $f(x) = a \cdot b^x$ where a and b are positive constants. The graph of $y = f(x)$ passes through the points $P = (1, 6)$ and $Q = (3, 54)$. A related function $g(x)$ is defined such that $g(x) = \log_b\left(\frac{f(x)}{a}\right)$. If the point R on the graph of $y = g(x)$ has an x-coordinate of 2, what is the y-coordinate of R?

a. 2

b. 3

c. $\log_3 9$

d. $\frac{1}{2}\log_3 18$

41 [Challenging] • Target: Even/odd properties and periodicity • Cross-Lesson: 9.03 The Binomial Theorem

In the xy-plane, an angle θ is in standard position. If the terminal side of θ passes through the point $(-5, 12)$, what is the value of $\sin(\theta)$?

a. $\frac{12}{13}$

b. $-\frac{5}{13}$

c. $-\frac{12}{13}$

d. $-\frac{12}{5}$

42 [Challenging] • Target: Solving basic trigonometric equations on the unit circle • Cross-Lesson: 1.01 Functions and Function Notation

The graph of the function $f(x) = A\sin(Bx + C) + D$ is shown below. The maximum value of the function is 5 and the minimum value is -1. The period of the function is $\frac{\pi}{2}$, and the graph passes through the point $(0, 2)$. Find the value of A.

a. 2

b. 3

c. 4

d. 6

43 [Challenging] • Target: Building the unit circle from special triangles • Cross-Lesson: 1.02 Domain and Range

Find all values of θ such that $\sin\theta, \cos\theta, \tan\theta$ are in arithmetic progression.

a. $\theta \approx 0.6435$ rad (a special angle satisfying $2\cos\theta = \sin\theta + \tan\theta$)

b. $\theta = \pi/3$

c. No solution exists

d. $\theta = \pi/4$

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44 [Challenging] • Target: Using the Pythagorean identity with unit circle points • Cross-Lesson: 1.03 Average Rate of Change

State the Pythagorean identity relating $\sec$ and $\tan$.

a. $\sec^2 x = \tan^2 x$

b. $\tan^2 x + 1 = \csc^2 x$

c. $\sec^2 x = 1 + \tan^2 x$

d. $\sec^2 x = 1 - \tan^2 x$

45 [Challenging] • Target: The unit circle picture • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Convert the rectangular point $(-1, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

a. $r = 1.414,\ \theta = 225°$

b. $r = 2.414,\ \theta = 135°$

c. $r = 1.414,\ \theta = 315°$

d. $r = 1.414,\ \theta = 135°$

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